

SAI SRI RAM MARNENI

AI Engineer | Generative AI & Agentic AI Systems | LLM Systems | Multi-Agent Orchestration | RAG Pipelines

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Visa: Stamp 2 (Student) → Stamp 1G Graduate Visa applied · Full work authorisation from Aug 2026

PROFESSIONAL SUMMARY

AI Engineer with 2 years of industry experience at Mphasis building production-grade Generative AI and Agentic AI systems — including LLM-powered automation, multi-agent orchestration frameworks, Retrieval-Augmented Generation (RAG) pipelines, and privacy-preserving NLP systems. Currently completing an MSc in Artificial Intelligence at Munster Technological University (graduating Aug 2026), with a thesis (SAMFO) addressing real-time schema drift detection and adaptive API failover in production agentic systems. Hands-on with Python, CrewAI, LangChain, FastAPI, and Model Context Protocol (MCP). Experienced in taking agentic AI systems from prototype to production, with a focus on runtime reliability, observability, and API integration. Targeting AI Engineer, LLM Engineer, and Generative AI Engineer roles in Ireland and Europe.

CORE TECHNICAL SKILLS

Generative AI & Agentic AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration, Prompt Engineering, Model Context Protocol (MCP), Schema Drift Detection, Runtime Observability, Agentic AI, NLP, LLM Fine-tuning

Frameworks & Libraries: CrewAI, LangChain, FastAPI, Streamlit, HTTPX, Scikit-learn, Pandas, NumPy

Languages: Python, SQL, JavaScript

Cloud & AI Services: Azure OpenAI, OpenAI API

Vector Databases & Storage: ChromaDB, PostgreSQL, MongoDB

DevOps & Tools: Git, Docker, GitHub Actions, CI/CD Pipelines, API Integration

PROFESSIONAL EXPERIENCE

Associate Software Engineer — AI & LLM Systems · Mphasis · Bengaluru, India | Aug 2023 – Aug 2025

5-Agent Supply Chain Automation System | CrewAI · Model Context Protocol (MCP) · PostgreSQL · Python

- Built and deployed a 5-agent supply chain automation system to production using CrewAI and Model Context Protocol (MCP), enabling AI-driven decision workflows across multi-turn enterprise interactions with SQL-backed data retrieval
- Coordinated 5 specialised agents handling inventory tracking, supplier optimisation, and transport cost analysis — reducing manual intervention in routine supply chain decisions
- Implemented MCP integration enabling structured, reliable tool calls between agents and the backend data layer, expanding the range of tasks handled autonomously

Employee Turnover Classification | Python · Scikit-learn · Pandas · SMOTE

- Improved employee turnover classification model accuracy from ~45% to ~78% through systematic ML engineering — EDA, SMOTE class balancing, k-fold cross-validation, and hyperparameter tuning
- Identified key drivers of employee attrition and surfaced actionable retention strategies from model outputs, enabling targeted HR interventions

Privacy-Preserving Content Moderation | LLaMA 2 · BERT · Mistral · GPT-3.5 · NLP

- Developed a real-time NLP-based content moderation system using 4 locally hosted LLMs (LLaMA 2, BERT, Mistral, GPT-3.5), eliminating reliance on third-party APIs and removing external data exposure risk for enterprise clients
- Enabled real-time harm and relevance assessment with zero third-party API dependency, laying the foundation for future plugin integrations

LLM Tooling & Pipeline Automation | Python · LangChain · FastAPI

- Engineered Python-based LLM solutions and internal AI tooling that automated data pipeline workflows, reducing manual processing effort and improving pipeline reliability across the engineering team

PROJECTS

SAMFO — Self-Adaptive Memory and Feedback Orchestrator | 2025–2026 | MSc Thesis · MTU

Python · CrewAI · FastAPI · HTTPX · Multi-Agent Systems · Agentic AI

- Designed and built a self-adaptive multi-agent orchestration framework coordinating 4–5 specialised agents with shared memory, runtime observability, and adaptive API reliability — addressing a core unsolved problem in production Generative AI systems

- Implemented real-time schema drift detection across agent-to-API communication layers, with rule-based failover logic that prevents failure propagation before it cascades to downstream agents
- Validated under controlled failure injection and live API conditions — demonstrated measurably improved resilience and observability compared to static orchestration baselines

Multi-Agent RAG & Contract Lifecycle Management (CLM) | 2024

Python · LangChain · OpenAI · CrewAI · ChromaDB · Streamlit

- Built a production-ready multi-agent Retrieval-Augmented Generation (RAG) pipeline enabling intelligent semantic search over 50–200 PDF contracts and enterprise documents using LangChain, OpenAI embeddings, and ChromaDB vector storage
- Optimised vector database retrieval through embedding evaluation and index tuning — zero-code Streamlit UI enabled non-technical users to query the document corpus directly

Multi-Agent Supply Chain Management System | 2024

Python · CrewAI · PostgreSQL · Model Context Protocol (MCP)

- Developed a 5-agent supply chain system with multi-turn interaction management, SQL-driven backend retrieval, and MCP integration for structured agent-to-tool communication

EDUCATION

MSc Artificial Intelligence · Munster Technological University (MTU), Cork, Ireland | Sep 2025 – Aug 2026 (expected)

Thesis: SAMFO — Self-Adaptive Memory and Feedback Orchestrator · Modules: Machine Learning, Deep Learning, NLP, Intelligent Systems, Data Analytics

B.Tech · QIS Institute of Technology, Andhra Pradesh, India | Jun 2018 – Jun 2022

GPA: 7.44/10

CERTIFICATIONS

AI-101 — Machine Learning, Deep Learning & Artificial Intelligence · Talent Next · Jan 2024

SQL and Database Management · NxtWave · Feb 2024

Full Stack Development · NxtWave · Jun 2023